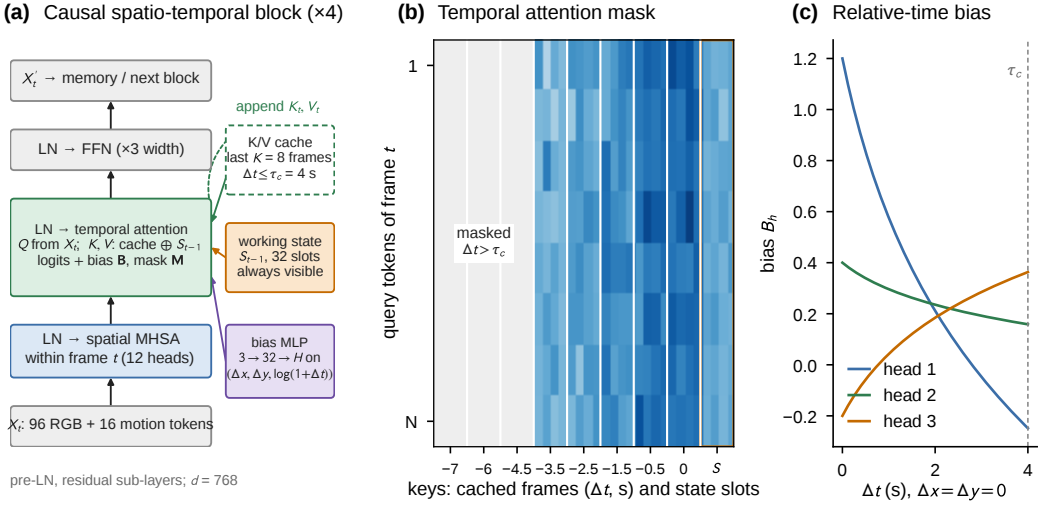
(a) Causal spatio-temporal block (×4)
X_t' → memory / next block
append K_t, V_t
LN → FFN (×3 width)
K/V cache
last K = 8 frames
$\Delta t \le \tau_c$ = 4 s
LN → temporal attention
Q from X_t'; K, V: cache ⊕ S_{t-1}
logits + bias $\mathbf{B}$, mask $\mathbf{M}$
working state
S_{t-1}, 32 slots
always visible
LN → spatial MHSA
within frame t (12 heads)
bias MLP
3 → 32 → H on
(Δx, Δy, log(1+Δt))
X_t: 96 RGB + 16 motion tokens
pre-LN, residual sub-layers; d = 768
(b) Temporal attention mask
query tokens of frame t
1
N
masked
$\Delta t > \tau_c$
−7 −6 −4.5 −3.5 −2.5 −1.5 −0.5 0 S
keys: cached frames (Δt, s) and state slots
(c) Relative-time bias
bias B_h
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
τ_c
head 1
head 2
head 3
0 2 4
Δt (s), $\Delta x = \Delta y = 0$